

A Wishart-column alternative to the inverse-Wishart LD relaxation

1 Setup

The notes in `susie-relax-report.tex` and `summary-susie-love-revisit.tex` use an inverse-Wishart prior on the latent LD matrix. This note considers the corresponding alternative

$$R \sim \mathcal{W}_J \left(\frac{\bar{R}}{\nu_0}, \nu_0 \right),$$

so that $\mathbb{E}(R) = \bar{R}$. Here $\mathcal{W}(V, \nu)$ denotes the Wishart distribution with scale matrix V and degrees of freedom ν . For a nonsingular $J \times J$ Wishart distribution, we need $\nu_0 > J - 1$. This is a practical difference from the inverse-Wishart parameterization, where ν_0 can be used more directly as a small positive relaxation parameter.

Throughout, $x \in \mathbb{R}^J$ is the vector of marginal summary statistics, N is the GWAS sample size, and $\bar{R}$ is a reference LD correlation matrix. For a fixed coordinate j , write

$$\mu_j = \bar{R}_{-j,j}, \quad \bar{S}_j = \bar{R}_{-j,-j} - \mu_j \mu_j^\top, \quad \Omega = \bar{R}^{-1}.$$

Let

$$d_j = R_{jj}, \quad z_j = \frac{R_{-j,j}}{R_{jj}}, \quad S_j^R = R_{-j,-j} - R_{-j,j} R_{jj}^{-1} R_{j,-j}.$$

The active column of R is $d_j u_j(z_j)$, where $u_j(z_j)$ is the vector with j -th entry equal to 1 and remaining entries equal to z_j .

2 Wishart-induced column distribution

By the standard block decomposition of a Wishart random matrix, after permuting coordinate j to the first position,

$$d_j \sim \text{Gamma} \left(\frac{\nu_0}{2}, \frac{\nu_0}{2} \right), \tag{1}$$

$$z_j \mid d_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{\bar{S}_j}{\nu_0 d_j} \right), \tag{2}$$

$$S_j^R \sim \mathcal{W}_{J-1} \left(\frac{\bar{S}_j}{\nu_0}, \nu_0 - 1 \right), \tag{3}$$

using the shape-rate parameterization for Gamma distributions. Moreover, S_j^R is independent of (d_j, z_j) . These facts imply

$$\mathbb{E}(d_j) = 1, \quad \text{Var}(d_j) = \frac{2}{\nu_0}, \quad \mathbb{E}(z_j \mid d_j) = \mu_j, \quad \text{Var}(z_j \mid d_j) = \frac{\bar{S}_j}{\nu_0 d_j}.$$

The mean of the Schur complement is

$$\mathbb{E}(S_j^R) = \frac{\nu_0 - 1}{\nu_0} \bar{S}_j.$$

This is not equal to $\bar{S}_j$, but the full matrix still has $\mathbb{E}(R) = \bar{R}$, because

$$\mathbb{E}\{d_j z_j z_j^\top\} = \mu_j \mu_j^\top + \frac{1}{\nu_0} \bar{S}_j.$$

Comparison with the inverse-Wishart column. Under the inverse-Wishart model used in the earlier notes, d_j is inverse-gamma and the relaxed column direction has a normal scale tied to the random Schur complement. Under the Wishart model, d_j is gamma and

$$z_j \mid d_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{\bar{S}_j}{\nu_0 d_j} \right),$$

so the column direction is tied to the reference Schur complement $\bar{S}_j$, not to S_j^R . This changes the conjugacy structure.

3 Fixed-variance SER-relax with a Wishart column

The closest analogue of the prototype in `susie-relax-report.tex` keeps the observation covariance fixed at $\bar{R}/N$, but replaces the inverse-Wishart column prior by the Wishart-induced column prior:

$$x \mid \beta, \gamma = j, d_j, z_j \sim \mathcal{N}_J \left(\beta d_j u_j(z_j), \frac{\bar{R}}{N} \right), \quad (4)$$

$$\beta \sim \mathcal{N}(0, \sigma_0^2), \quad (5)$$

$$\gamma \sim \text{Unif}\{1, \dots, J\}, \quad (6)$$

$$d_j \sim \text{Gamma} \left(\frac{\nu_0}{2}, \frac{\nu_0}{2} \right), \quad (7)$$

$$z_j \mid d_j \sim \mathcal{N}_{J-1} \left(\mu_j, \frac{\bar{S}_j}{\nu_0 d_j} \right). \quad (8)$$

This model is computationally close to the inverse-Wishart prototype, but the factor $q_j(d_j)$ changes because d_j appears in the likelihood and in the precision of the z_j prior.

Integrating out z_1 . For concreteness, take $\gamma = 1$ and write

$$x = \begin{pmatrix} x_1 \\ x_{-1} \end{pmatrix}, \quad \bar{R} = \begin{pmatrix} 1 & \mu_1^\top \\ \mu_1 & \bar{S}_1 + \mu_1 \mu_1^\top \end{pmatrix}.$$

Conditional on (β, d_1) , the only random part of the mean is $\beta d_1 z_1$, and

$$z_1 \mid d_1 \sim \mathcal{N}_{J-1} \left(\mu_1, \frac{\bar{S}_1}{\nu_0 d_1} \right).$$

Thus, after integrating over z_1 ,

$$x \mid \beta, d_1, \gamma = 1 \sim \mathcal{N}_J \left(\beta d_1 \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix}, \frac{\bar{R}}{N} + \frac{\beta^2 d_1}{\nu_0} \begin{pmatrix} 0 & 0 \\ 0 & \bar{S}_1 \end{pmatrix} \right).$$

Equivalently, using the block normal conditional,

$$\begin{aligned} p(x \mid \beta, d_1, \gamma = 1) &= \mathcal{N} \left(x_1 \mid \beta d_1, \frac{1}{N} \right) \\ &\times \mathcal{N}_{J-1} \left(x_{-1} \mid x_1 \mu_1, \left(\frac{1}{N} + \frac{\beta^2 d_1}{\nu_0} \right) \bar{S}_1 \right). \end{aligned} \quad (9)$$

This likelihood keeps the same conditional-mean simplification: after conditioning on x_1 , the mean of x_{-1} is $x_1 \mu_1$ and does not involve β or d_1 .

Now write the same likelihood explicitly in terms of

$$q = x^\top \bar{R}^{-1} x, \quad \zeta = \frac{N \beta^2 d_1}{\nu_0}.$$

The covariance matrix above is $N^{-1} \tilde{R}$, where

$$\tilde{R} = \bar{R} + \zeta \begin{pmatrix} 0 & 0 \\ 0 & \bar{S}_1 \end{pmatrix}.$$

By the block determinant identity,

$$|\tilde{R}| = (1 + \zeta)^{J-1} |\bar{R}|,$$

and by the block inverse identity,

$$x^\top \tilde{R}^{-1} x = \frac{1}{1 + \zeta} x^\top \bar{R}^{-1} x + \frac{\zeta}{1 + \zeta} x_1^2.$$

Also, since $(1, \mu_1^\top)^\top$ is the first column of $\tilde{R}$,

$$\tilde{R}^{-1} \begin{pmatrix} 1 \\ \mu_1 \end{pmatrix} = e_1.$$

Therefore

$$\begin{aligned} p(x \mid \beta, d_1, \gamma = 1) &= \frac{N^{J/2}}{(2\pi)^{J/2} |\bar{R}|^{1/2} (1 + \zeta)^{(J-1)/2}} \\ &\times \exp \left\{ -\frac{N}{2} \left[\frac{q}{1 + \zeta} + \frac{\zeta}{1 + \zeta} x_1^2 - 2\beta d_1 x_1 + \beta^2 d_1^2 \right] \right\}. \end{aligned} \quad (10)$$

Equivalently,

$$\begin{aligned} p(x \mid \beta, d_1, \gamma = 1) &\propto (1 + \zeta)^{-(J-1)/2} \exp \left\{ -\frac{Nq}{2(1 + \zeta)} \right\} \\ &\times \exp \left\{ -\frac{N\zeta x_1^2}{2(1 + \zeta)} \right\} \exp \left\{ N\beta d_1 x_1 - \frac{N}{2} \beta^2 d_1^2 \right\}, \end{aligned} \quad (11)$$

where the omitted proportionality constant is common across coordinates.

3.1 Mean-field updates

Use

$$q(\gamma = j, \beta, d_j, z_j) = \pi_j q_j(\beta) q_j(d_j) q_j(z_j),$$

where

$$q_j(\beta) = \mathcal{N}(m_{\beta j}, v_{\beta j}), \quad q_j(z_j) = \mathcal{N}_{J-1}(m_{zj}, V_{zj}).$$

Let

$$\bar{\beta}_j = m_{\beta j}, \quad \overline{\beta^2}_j = v_{\beta j} + m_{\beta j}^2, \quad \bar{d}_j = \mathbb{E}_j(d_j), \quad \overline{d^2}_j = \mathbb{E}_j(d_j^2).$$

Define

$$A_j = 1 + \text{tr}(\bar{S}_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top \bar{S}_j^{-1} (m_{zj} - \mu_j), \quad (12)$$

$$B_j = x_j + (m_{zj} - \mu_j)^\top \bar{S}_j^{-1} (x_{-j} - \mu_j x_j), \quad (13)$$

$$Q_j = \text{tr}(\bar{S}_j^{-1} V_{zj}) + (m_{zj} - \mu_j)^\top \bar{S}_j^{-1} (m_{zj} - \mu_j). \quad (14)$$

The Gaussian update for the effect size is

$$v_{\beta j} = \left(\sigma_0^{-2} + N \overline{d^2}_j A_j \right)^{-1}, \quad (15)$$

$$m_{\beta j} = v_{\beta j} N \bar{d}_j B_j. \quad (16)$$

The Gaussian update for the column direction is

$$V_{zj} = \frac{\bar{S}_j}{N \overline{\beta^2}_j \overline{d^2}_j + \nu_0 \bar{d}_j}, \quad (17)$$

$$m_{zj} = \mu_j + \frac{N \bar{\beta}_j \bar{d}_j}{N \overline{\beta^2}_j \overline{d^2}_j + \nu_0 \bar{d}_j} (x_{-j} - \mu_j x_j). \quad (18)$$

The optimal density for d_j is non-standard:

$$q_j(d_j) \propto d_j^{\nu_0/2-1+(J-1)/2} \exp \left\{ - \left(\frac{\nu_0}{2} + \frac{\nu_0 Q_j}{2} \right) d_j - \frac{N}{2} \overline{\beta^2}_j A_j d_j^2 + N \bar{\beta}_j B_j d_j \right\}, \quad d_j > 0.$$

In terms of $r_j = \log d_j$, including the Jacobian,

$$\ell_j(r) = \left(\frac{\nu_0}{2} - 1 + \frac{J-1}{2} \right) r - \left(\frac{\nu_0}{2} + \frac{\nu_0 Q_j}{2} \right) e^r - \frac{N}{2} \overline{\beta^2}_j A_j e^{2r} + N \bar{\beta}_j B_j e^r + r.$$

One-dimensional quadrature over r gives $\mathbb{E}_j(d_j)$, $\mathbb{E}_j(d_j^2)$, $\mathbb{E}_j(\log d_j)$, and $E_j \log q_j(d_j)$.

3.2 Expected log terms

The expected log-likelihood has the same form as in the inverse-Wishart prototype:

$$\begin{aligned} E_j \log p(x \mid \beta, d_j, z_j, \gamma = j) &= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \log \left| \frac{\bar{R}}{N} \right| \\ &\quad - \frac{N}{2} \left[x^\top \Omega x - 2 \bar{\beta}_j \bar{d}_j B_j + \overline{\beta^2}_j \overline{d^2}_j A_j \right]. \end{aligned} \quad (19)$$

The Wishart-column prior contributions are

$$\begin{aligned}
E_j \log p(d_j \mid \nu_0) &= \frac{\nu_0}{2} \log \frac{\nu_0}{2} - \log \Gamma\left(\frac{\nu_0}{2}\right) + \left(\frac{\nu_0}{2} - 1\right) \mathbb{E}_j \log d_j - \frac{\nu_0}{2} \mathbb{E}_j d_j, \\
E_j \log p(z_j \mid d_j, \nu_0) &= -\frac{J-1}{2} \log(2\pi) - \frac{1}{2} \log |\bar{S}_j| + \frac{J-1}{2} \log \nu_0 + \frac{J-1}{2} \mathbb{E}_j \log d_j - \frac{\nu_0}{2} \mathbb{E}_j d_j Q_j.
\end{aligned} \tag{20}$$

(21)

These terms can be substituted into the usual local lower bound and used to update the coordinate probabilities π_j .

4 Takeaways

1. A Wishart prior $R \sim \mathcal{W}(\bar{R}/\nu_0, \nu_0)$ induces

$$d_j \sim \text{Gamma}(\nu_0/2, \nu_0/2), \quad z_j \mid d_j \sim \mathcal{N}_{J-1}(\mu_j, \bar{S}_j/(\nu_0 d_j)).$$

2. In the fixed-variance SER-relax analogue, variational inference is nearly the same as the inverse-Wishart prototype, except $q_j(d_j)$ has gamma-type behavior near zero and the z_j prior contributes $\nu_0 d_j Q_j/2$.
3. After integrating out z_1 in the fixed-variance model, the likelihood factorizes as

$$p(x \mid \beta, d_1, \gamma = 1) = \mathcal{N}\left(x_1 \mid \beta d_1, \frac{1}{N}\right) \mathcal{N}_{J-1}\left(x_{-1} \mid x_1 \mu_1, \left(\frac{1}{N} + \frac{\beta^2 d_1}{\nu_0}\right) \bar{S}_1\right).$$